The Great Plains Laboratory, Inc.

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[redacted]
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Public copy: a private street address is removed. Otherwise unchanged.

The Great Plains Laboratory, Inc.

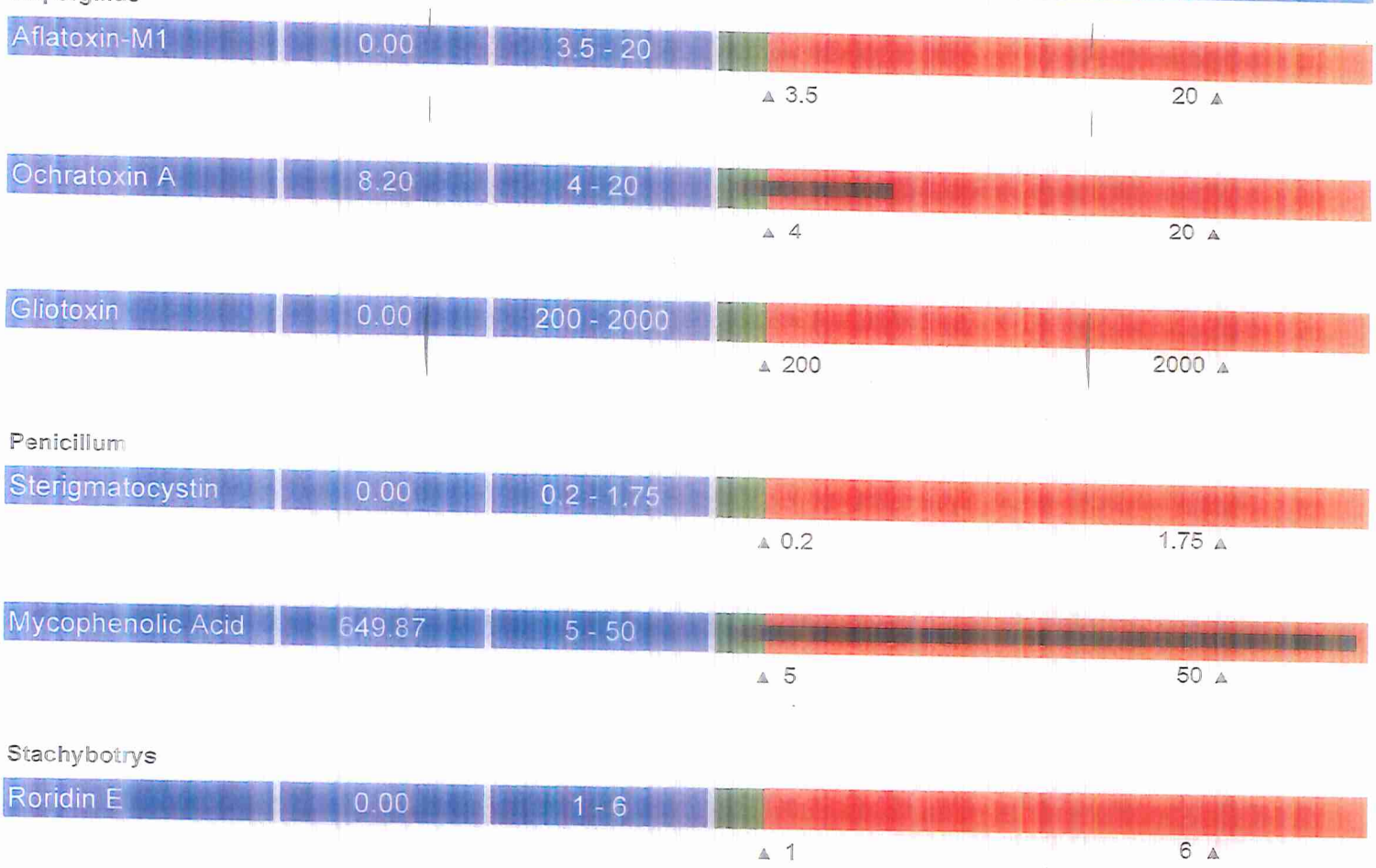
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GPL-MYCOTOX

Requisition #	714255	Physician Name	AHVIE HERSKOWITZ
Patient Name	Stephen Russell	Date of Collection	7/31/2019
Patient BirthDate	12/4/1973	Time of Collection	3:00 PM
Sex	M	Print Date	8/6/2019

MycoTox Profile

Metabolite	Results (ng/g creatinine)	Common Range of Positive Results
Aspergillus		
Aflatoxin-M1	0.00	3.5 - 20
Ochratoxin A	8.20	4 - 20
Gliotoxin	0.00	200 - 2000
Penicillum		
Sterigmatocystin	0.00	0.2 - 1.75
Mycophenolic Acid	649.87	5 - 50
Stachybotrys		
Roridin E	0.00	1 - 6



Testing performed by The Great Plains Laboratory, Inc., Lenexa, Kansas. The Great Plains Laboratory has developed and determined the performance characteristics of this test. The test has not been evaluated by the U.S. Food and Drug Administration. The FDA does not currently regulate such testing.

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Patient Name: Stephen Russell Requisition #: 714255

GPL-MYCOTOX

Metabolite	Results (ng/g creatinine)	Common Range of Positive Results
Verrucarin A	0.00	1 - 10

▲ 1 10 ▲

Fusarium

Enniatin B	0.00	0.07 - 1

▲ 0.07 1 ▲

Zearalenone	0.00	0.5 - 10

▲ 0.5 10 ▲

Chaetomium globosum

Chaetoglobosin A	0.00	20 - 80

▲ 20 80 ▲

Multiple Mold Species

Citrinin	< 10.00	10 - 50

▲ 10 50 ▲

Ochratoxin: Ochratoxin A (OTA) is a nephrotoxic, immunotoxic, and carcinogenic mycotoxin. This chemical is produced by molds in the Aspergillus and Penicillium families. Exposure is done primarily through water damaged buildings. Minimal exposure can occur through contaminated foods such as cereals, grape juices, dairy, spices, wine, dried vine fruit, and coffee. Exposure to OTA can also come from inhalation exposure in water-damaged buildings. OTA can lead to kidney disease and adverse neurological effects. Studies have shown that OTA can lead to significant oxidative damage to multiple brain regions and is highly nephrotoxic. Dopamine levels in the brain of mice have been shown to be decreased after exposure to OTA. Some studies have hypothesized that OTA may contribute to the development of neurodegenerative diseases such as Alzheimer's and Parkinson's. Treatment should be aimed at removing the source of exposure. Agents such as oral cholestyramine, charcoal, and phenylalanine can help prevent the absorption of these toxins from food. Antioxidants such as vitamins A, E, C, NAC, rosmarinic acid, and liposomal glutathione alone or in combination have been shown to mitigate the oxidative effects of the toxin. Bentonite or zeolite clay is reported to reduce the absorption of multiple mycotoxins found in food, including OTA. Studies have also shown that OTA is present in sweat, which supports the use of sauna as a treatment to increase the excretion of OTA. (PMID 17195275, 16621780, 16293235, 27521635, 22069626, 24792326, 22253638, 16140385, 2467220, 16844142, 19148691, 22069658, 16019795, 18286403, 15781206, 11439224, 17092826, 32710148)

Mycophenolic Acid: Mycophenolic Acid (MPA) produced by the Penicillium fungus. MPA is an immunosuppressant which inhibits the proliferation of B and T lymphocytes. MPA exposure can increase the risk of opportunistic infections such as Clostridia and Candida. MPA is associated with miscarriage and congenital malformations when the woman is exposed in pregnancy. (PMID: 28646113, 27809954, 27599910)